EXPERIMENTER RUN SHEET

Follow My Voice

Experimenter protocol / October 20, 2026

SAY / ASK Participant-facing speech is set in dark upright type.

ACTION / RESEARCHER *Experimenter-only instructions are italic and muted gold-brown. Do not read them aloud.*

Participant ID _____ **Run** _____

Date _____ **Researcher** _____

Build _____ **Voice order** _____

Preferred voice ID _____ **Survey version** _____

CURRENT DESIGN / RESEARCHER ONLY *Two counterbalanced voice blocks: participant-preferred catalog voice and self-similar synthetic voice. Each block will contain 40 measured trials (80 total); one practice trial will be separate. Use the saved assigned order. Each trial will be one hidden-target search. Both conditions will use the same gaze-contingent guidance policy and right-controller selection. Say "first block" and "second block" without suggesting either voice will improve performance.*

Before participant arrives

- *Sanitize and test the HTC Vive Focus Vision and right controller. Check the rotating chair and clear the seated movement area.*
- *Open the frozen study build and the versioned questionnaires. Confirm the build ID, audio-library version, recording destination and available storage.*
- *Enter coded participant/run IDs and saved voice-block order. Verify one practice trial and 40 measured trials per block. Confirm researcher visual aids and passthrough are set consistently across conditions.*
- *Prepare consent, baseline survey, both block surveys, final interview and session/event record. Use one defined TLX collection route and avoid duplicate administration.*

1 Welcome and consent

SAY "Thank you for coming. Today you will be performing a search task in mixed reality. In this task, you will stay seated and turn to find virtual objects. A voice assistant will know the target and will guide you toward it without naming its color or shape. You will be performing this task for a number of trials over two task blocks, each using a different assistant voice. You may pause, take a break or stop at any time without penalty."

SAY "We will ask you to record a short voice sample to create a synthetic version of your voice. We will also record eye-gaze and task data under a study code. Please read the consent form for what will be collected, where it will be processed and how it will be handled."

ACTION *Obtain consent.*

2 Initial Qualtrics and Voice Setup

SAY "Please complete the background and current-comfort questions before we begin. Use only the study code already entered. Do not add your name or contact details. You may skip optional questions. We will also ask what kind of assistant voice you would prefer. There are no right or wrong preferences."

ACTION *Confirm the coded participant and run IDs. Include the baseline demographics and current-comfort items, followed by the separate questions below. Keep skipped responses distinct from "no preference."*

SURVEY ITEM / AGENT GENDER PREFERENCE "What gender presentation would you prefer for the agent's voice?"

ACTION / CANDIDATE PREPARATION *Use the frozen candidate catalog and common preview text. Have the participant preview voices, choose one and confirm the selection. Save requested gender presentation separately from actual voice ID; record unavailable requests and "no preference" distinctly. Do not infer or match the agent's gender from participant gender. Record the configured catalog and presentation order.*

VOICE RATINGS *During setup, rate each auditioned voice, including the self-similar voice, from 0 (Do not like at all) to 100 (Like extremely). Record voice ID and audition order. At the final survey, collect liking again separately for each voice block and the 1–7 voice-similarity item with Strongly disagree / Strongly agree endpoints. Keep setup and final retrospective responses separate.*

SAY "We will now record the short voice sample described in the consent form. After the tone, please read the displayed passage in your usual speaking voice."

ASK / AUDIO CHECK "Can you hear the voice clearly and at a comfortable volume?"

3 Fit and calibrate the headset

SAY "Remain seated in this chair. You may turn the chair, your body and your head to look around, but do not stand or walk. Tell me immediately if you feel dizzy, nauseated, strained, uncomfortable or unsteady."

ACTION *Fit the headset, verify both eyes' tracking and freshness/validity feedback, and perform the headset's calibration. Then use independent validation targets across low/middle/high positions, wall boundaries and relevant distances. Record the validation result. Revalidate after headset adjustment and between blocks.*

OPEN DECISION / RELEASE GATE *Freeze numerical validation/drift criteria, retry limits and a tracking-failure procedure using headset/task pilots before confirmatory collection. Head direction must not silently replace missing eye gaze. Ray intersections are not validated fixations, and polling at 120 Hz does not establish distinct 120 Hz observations.*

ASK "Can you see clearly? Does the equipment feel comfortable? Do you feel well enough to continue?"

4. Pre-Block Activity

SAY "First, I will explain the task. In each trial, the agent will know which object you need to find. It will guide you toward the object without telling you its color or shape in advance. You will use the agent's directions to reach the correct area and then make smaller adjustments up, down or to either side."

SAY "You will point the right controller at your choice to highlight it, then press the right trigger to select it. Looking alone will not select an object. Selecting the correct object will complete the trial. If you select an incorrect object, you will continue searching for the same target."

SAY "Before each search, you will look at the readiness cross until it charges. You will hear the first direction before the objects appear. When they appear, the timed search will begin. You will not need to select a room surface."

SAY "If you would like a rest, let me know. We will normally take it between trials, before the next search begins. You can also ask us to pause or stop immediately at any time."

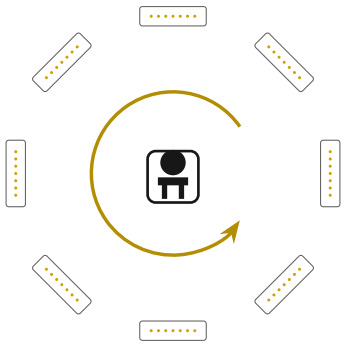
BREAKS AND TIMING *Requested rest breaks normally occur after a trial ends and before the next search begins. Do not start the next search until the participant is ready. Breaks and questionnaires are outside active-search time. If a participant needs to pause during a trial, pause immediately and record the interruption and its duration separately. Verify that the frozen build excludes paused time from active search; if it cannot, mark that trial interrupted and retain the raw timing for the prespecified analysis. Retain the interrupted-trial record; do not silently replace it or count break time as search time.*

5. Familiarization

ACTION *Seat the participant at the intended center and capture the designated room-front reference using the right-controller start control. Do not use keyboard shortcuts in participant instructions. Keep the same room reference across the session. The readiness cross for later trials will follow the saved schedule's starting reference, currently the previous target wall; it will not require a fixed front direction on every trial. Verify this in the frozen build.*

SAY "Objects will appear all around you. You will use the chair to turn while staying seated and follow the agent's directions. You will point with your right controller and press its trigger when you choose an object."

Objects will appear around you



Remain seated. Turn to look around.

Schematic, not to scale. Illustrates seated rotation only.

SAY "The next trial will be a practice trial to help you become familiar with the task."

ACTION *Complete one practice trial before measured collection. Verify understanding of the cross, guidance and right-trigger selection. Record the practice voice and any extra explanation or repeated practice. The single practice voice assignment remains to be specified consistently before collection. Apply the same predefined proficiency rule to every participant; freeze that rule before collection.*

ASK "Do you have any questions about the controls or task?"

SAY "When you are ready, we will begin the first task block."

6 First task block

PLANNED SCHEDULE *40 measured trials in this voice block; the practice trial is separate. Verify the configured count before participant use.*

RESEARCHER SETTINGS *Confirm the saved first voice and planned trial schedule. Keep object layout, selection method, guidance policy and the finalized prompt wording constant across voices.*

SAY "We will now begin the first task block. Find each target as quickly and accurately as you comfortably can. Stay seated and use the same right-controller ray and trigger selection you practiced. Tell me if you need to pause."

ACTION *Start block; monitor safety.*

AT COMPLETION SAY "That task block is complete. Please stop and lower your hands comfortably."

SAY "You may take a break now. Please tell me when you feel ready to continue."

ACTION *Record break duration and current comfort. Recheck headset fit and independently validate tracking before the next block; recalibrate if required by the frozen acceptance rules.*

BEFORE CONTINUING ASK "Do you feel comfortable and well enough to continue?"

7 Second task block

PLANNED SCHEDULE *40 measured trials using the other assigned voice. Retain incomplete or interrupted trials with their outcomes; do not add replacement trials merely to reach 40 valid attempts.*

SAY "We will now begin the second task block. This block will use the other assistant voice. The task and selection method will remain the same. Find each target as quickly and accurately as you comfortably can, and tell me if you need to pause."

ACTION *Start block; monitor safety.*

AT COMPLETION SAY "The search task is complete. Please stop turning. I will help you remove the headset, then you will complete the final questionnaires about both blocks."

8 Final questionnaire

SAY "Please complete the final questionnaires. You will answer one set about the first block and another about the second block. Each set will identify which voice you used. Please answer about the indicated block, rather than combining both experiences."

SAY "Some follow-up questions may appear depending on your answers about the two voices. Answer based on your own experience and tell me when you have submitted it."

EXPLORATORY FOLLOW-UP *After open responses, ask "Did either voice feel like your own thoughts, like another person helping, or neither?" Record this as an exploratory report, not proof of an internal-dialogue mechanism.*

ACTION *After headset removal, administer the post-exposure SSQ and safety check first. Then administer the two condition-specific sets in block order: six workload ratings, 24 IMI items, 21 selected Guo items and 14 custom numeric ratings per set, plus technical checks and optional text. Use identical item wording and order. Preserve the referenced block/voice ID separately from the actual final-survey timestamp. Finish with comparative preference and open debrief questions. Use the balanced follow-up guide only after open responses, avoiding duplicate questions.*

9 Debrief and finish

SAY "Thank you. This study compares participant-preferred and self-similar voices during gaze-guided object search. Both voices provide gaze-contingent assistance. We are examining task performance and how participants experience the activity and assistant."

SAY "Your data are associated with a coded participant ID. Your voice may still be recognizable; it will be handled as described in the consent form. Please avoid discussing the voice conditions or study details with people who may participate later, because knowing them in advance could affect their behavior."

ASK "Do you have any questions about the study or your experience?"

SAY "Thank you for participating. We are finished."

Immediate stop or pause criteria

PAUSE OR STOP

Participant request, dizziness, nausea, headache, eye strain, fatigue, pain, distress, unsafe movement or equipment failure. Help the participant remain seated and remove the headset when needed.

Final researcher check

- *Confirm trial counts and practice labels; retain incomplete/interrupted trials and reasons. Match participant, run, block, trial, voice and build identifiers.*

- *Confirm the per-trial performance/subjective CSV and separate time-series motion/gaze/event/replay data. Check raw timestamps, validity, target/object positions and survey joins; distinguish missing values from zero.*
- *Preserve object-reveal, first target-gaze intersection, qualified gaze confirmation, first target-controller intersection, controller confirmation, selection attempts and correct-trigger timestamps as distinct events. Do not label one as another.*
- *Record completion time, starting theta, signed height delta, target distance, instruction count, incorrect selections and interruption duration. Repeat block-level survey values on trial rows only with scope/block labels; they are not independent trial responses.*
- *Confirm both condition-specific sets of the final questionnaire and the interview/comfort records, or why incomplete. Secure exports and voice assets according to the approved retention plan; sanitize equipment.*

AUDIO AND READINESS CONFIGURATION *Archive the output device, volume/spatialization settings, microphone, recording passage, synthesis model and audio-library version. Freeze the cross charging duration and validity region, object collider/vicinity definitions and incorrect-selection feedback; apply the same settings across voice conditions.*

Difficulty and timing record / researcher only

CONFIRMED OUTCOME *Active trial completion time will be the primary outcome, accounting for horizontal theta toward the target from the starting reference. Record the opening cue separately: it will precede object reveal and the active-search timer. Completion will be the correct right-trigger selection.*

OPENING-CUE CHECK *Log cue onset/end, object reveal and head pose at cue onset and reveal. Participants may turn before the timer starts. Proposed sensitivity outcome: cue-onset-to-correct-selection time, alongside the primary active-search time. Starting theta alone will not account for movement during the cue.*

THETA *Keep nominal wall-angle category, exact starting target bearing, and actual head rotation distinct. The schedule will allocate eight trials at each absolute nominal angle (0, 45, 90, 135 and 180 degrees) per block in a randomized order. Save signs, positions, reference definition, seed and realized schedule. Calculate pre-cue theta from the horizontal origin-to-cross and origin-to-target vectors. Retain the sign and absolute angle; do not divide completion time by degrees.*

CONFIRMED HEIGHT REFERENCE *Signed height delta will be target-center world Y minus readiness-cross-center world Y, in metres. Positive means above the cross; negative means below. Save both positions. The intended quota is 20 above and 20 below per block; verify implementation and freeze feasible ranges/deadband before claiming the quota is met.*

BEFORE CONFIRMATORY RECRUITMENT

Complete the theta-adjusted power analysis and recruitment plan; set tracking acceptance, practice proficiency, timeout, interruption and missing-data rules; verify the vertical schedule and acoustic checks; reconcile consent with actual service processing and retention.

Appendix: Self-similar voice recording passage

ACTION *Use the recording passage displayed by the frozen build and archive its version. The text below is the retained passage from the source run sheet; verify it matches the release before use. Its color/shape examples are recording material, not target instructions during measured trials. Read in a usual, natural voice without deliberately adding a whisper or inner-thought style.*

CURRENT RECORDING PASSAGE

"Locate the red sphere. Look toward the upper shelf. Move your gaze slowly to the right. Check the blue cube near the center. Compare each object's color and shape. Scan the lower row from left to right. Look beside the purple cylinder. Point the controller ray at the target and check its highlight. Press the trigger to select the matching object. Take a short pause. Get ready for the next trial."